

MC60 OpenCPU

GS3 SDK — Third Test Report (GPS Fix)

SDK MC60_OpenCPU_GS3_SDK_V1.8

Toolchain Arm GNU Toolchain 15.2.1

Host OS Windows 11

Date July 2026

Example GPS (GNSS) — Live Fix

Interface PuTTY via CP2102 (COM6, 115200 baud)

Internal documentation — not for distribution

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1. Introduction

This report documents the third test of the Quectel MC60 module using the MC60_OpenCPU_GS3_SDK_V1.8. The first test established the cross-compilation toolchain, build workflow, and confirmed that the GPS example executes and emits NMEA sentences over the UART — but at that stage the GNSS receiver had not yet acquired satellites (status V, Void), as the module was tested indoors without a clear sky view. The second test switched to the Bluetooth example and verified that the Bluetooth radio, device scan, and OpenCPU BT API all functioned correctly.

The objective of this third test was to return to the GPS example, wait for the GNSS receiver to acquire enough satellites to produce a valid fix outdoors, and then verify the reported coordinates against a known map location. The firmware, toolchain, and flash procedure were unchanged from the first two tests.

2. Firmware and Setup

No changes to the firmware source or build environment were required for this test. The GPS example binary produced during the first test — compiled with `C_PREDEF=-D __CUSTOMER_CODE__ -D __EXAMPLE_GPS__` using the Arm GNU Toolchain 15.2.1 — was already present on the module. The same QFlash V4.0 configuration (SCAT, DA, APPGS3MDM32A01.bin) and PuTTY serial monitor settings (COM6, 115200 baud) from the first test were reused without modification.

Reused configuration from First Test

Firmware binary	APPGS3MDM32A01.bin (GPS example, GCC 15.2.1)
Flash tool	QFlash V4.0 – COM6, MC60, PASS confirmed
Serial monitor	PuTTY – Serial, COM6, 115200 baud
Preprocessor flag	C_PREDEF=-D __CUSTOMER_CODE__ -D __EXAMPLE_GPS__

Figure 1 shows the development environment at the time of the Bluetooth test (second test), included here for reference, as it also shows QFlash reporting **PASS, FW upgrade success** with the same scatter, DA, and binary files that remained on the module for this third test.

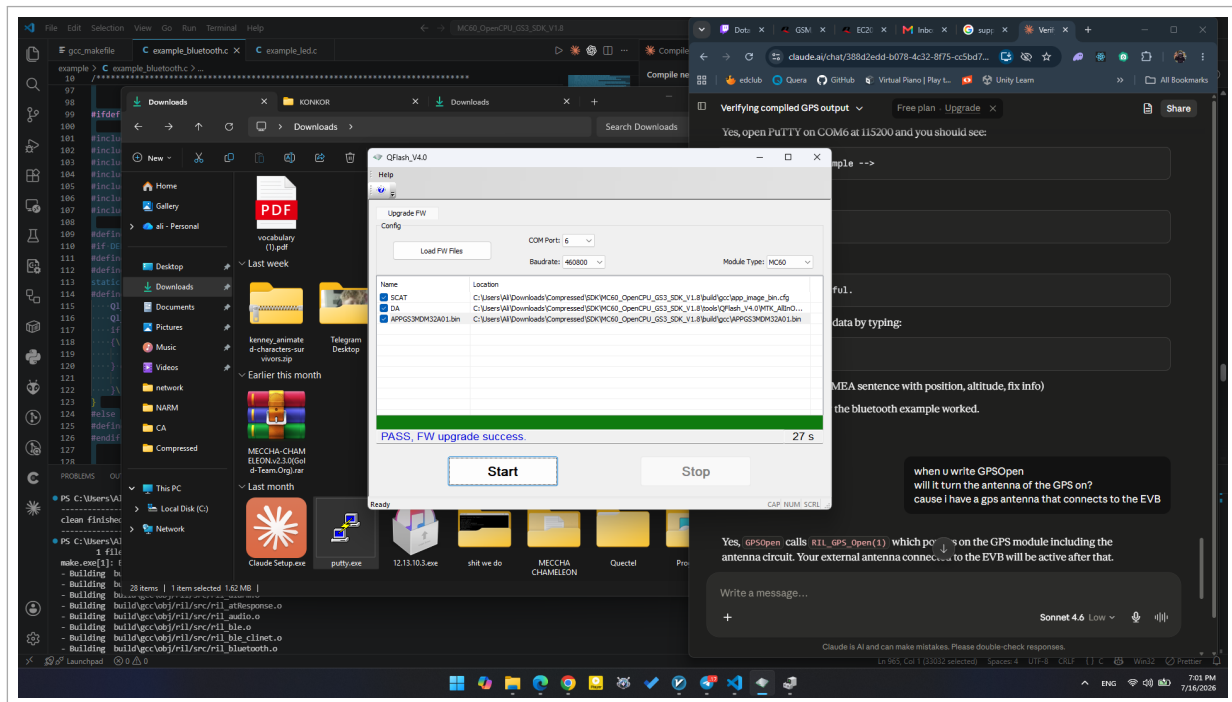


Figure 1: Development environment during the second test showing QFlash V4.0 reporting PASS (27 s). The same firmware binary and flash configuration were used for this third GPS test.

3. Waiting for a Valid Fix

After powering on the module, the GNSS subsystem immediately began emitting NMEA sentences, as observed in the first test. The initial sentences showed a status of V (Void) and empty coordinate fields, meaning no fix had yet been acquired. This is normal behaviour on a cold start: the receiver must download almanac and ephemeris data from satellites before it can compute a position.

The two early sentences below are representative of the pre-fix output:

```
+QGNSSRD: $GNGGA,000000.094,,,,,0,0,,M,,M,,*5B
+QGNSSRD: $GNGGA,153449.091,,,,,0,0,,M,,M,,*50
```

In both cases the coordinate fields (latitude, longitude, altitude) are empty and the fix quality indicator is 0, confirming that no valid position was available. After moving the antenna to a position with a clear line of sight to the sky and waiting approximately one to two minutes, the receiver acquired satellites and began

producing valid fixes.

Cold-start TTFF note: Time-To-First-Fix on a cold start (no cached almanac) is typically 30–90 seconds outdoors. The module used the external patch antenna connected to the EVB, which provided sufficient signal once positioned near a window with unobstructed sky view.

4. GNSS Output and Analysis

4.1. Terminal Output

Once a fix was acquired, the module began emitting \$GNGGA sentences containing valid coordinate data. Figure 2 shows the full PuTTY terminal capture, spanning the pre-fix period through to the first confirmed valid fix sentences.

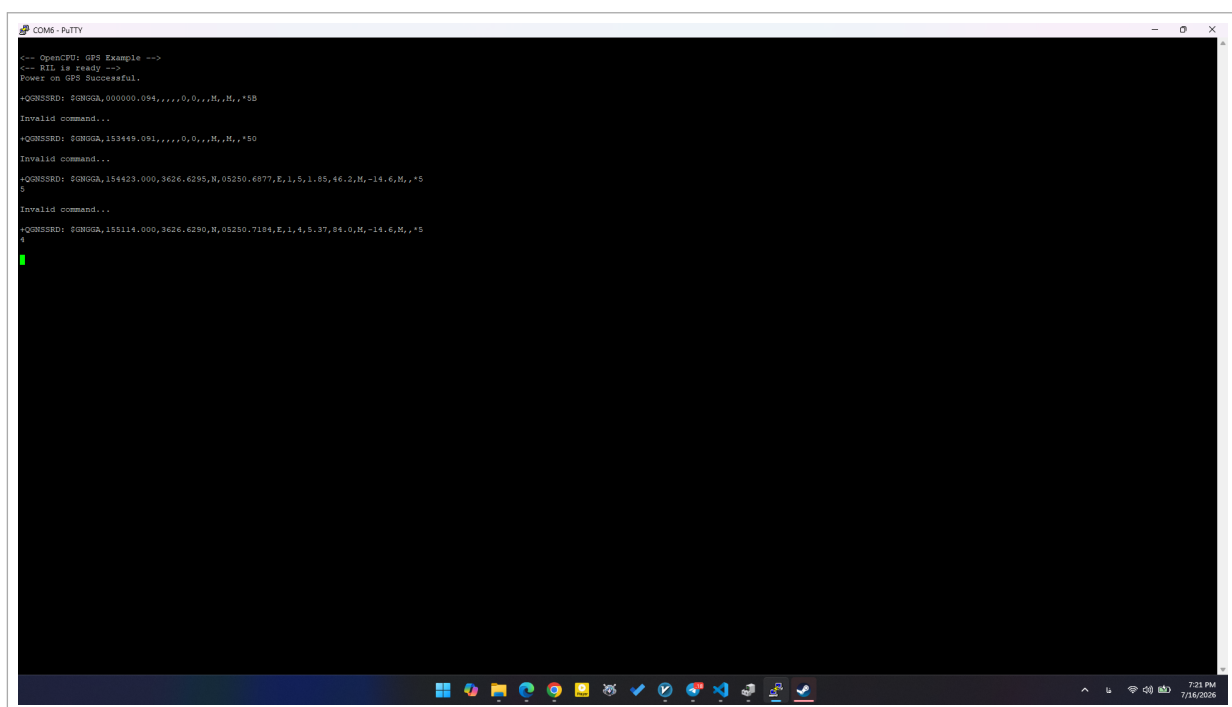


Figure 2: PuTTY terminal output (COM6, 115200 baud) showing the progression from no-fix (empty coordinate fields, quality = 0) to valid \$GNGGA sentences with coordinates, 4–5 satellites, and altitude.

4.2. NMEA Sentence Analysis

The module outputs +QGNSSRD URC (Unsolicited Result Code) messages wrapping standard NMEA \$GNGGA sentences. Once a fix was obtained, the sentences contained valid position data. The two fix sentences captured are:

```

+QGNSSRD: $GNGGA,154423.000,3626.6295,N,05250.6877,E,1,5,1.85,46.2,M,-14.6,M,,*5
+QGNSSRD: $GNGGA,155114.000,3626.6290,N,05250.7184,E,1,4,5.37,84.0,M,-14.6,M,,*5

```

Each field in the first fix sentence is decoded in Table 1:

Table 1: \$GNGGA field breakdown — first valid fix (154423.000 UTC)

Field	Value	Meaning
Time	154423.000	15:44:23 UTC
Latitude	3626.6295, N	36°26.6295' North → 36.44382° N
Longitude	05250.6877, E	52°50.6877' East → 52.84479° E
Fix quality	1	GPS fix acquired (1 = valid)
Satellites in use	5	Five satellites tracked
HDOP	1.85	Horizontal dilution of precision
Altitude	46.2 M	46.2 metres above mean sea level
Geoid separation	-14.6 M	WGS84 geoid correction for the area

4.3. Second Fix Sentence

A second valid fix was captured approximately 41 seconds later, showing a slightly refined position and a reduced satellite count (4 vs 5) but still a valid fix (quality = 1):

```
+QGNSSRD: $GNGGA,155114.000,3626.6290,N,05250.7184,E,1,4,5.37,84.0,M,-14.6,M,,*5
```

Note on altitude discrepancy: The altitude in the second sentence (84.0 m) differs significantly from the first (46.2 m). This is attributable to the higher HDOP of 5.37 in the second reading (versus 1.85 in the first), indicating reduced geometric accuracy when fewer well-positioned satellites were available. The first fix (HDOP 1.85, 5 satellites) is the more reliable of the two readings.

5. Location Verification

To verify the reported coordinates, the decimal-degree equivalent was derived from the NMEA degree-minute format and cross-checked against Google Maps.

The NMEA-to-decimal conversion for the first fix is straightforward:

```
Latitude: 3626.6295 N -> 36 + (26.6295 / 60) = 36.44382 deg N
Longitude: 05250.6877 E -> 52 + (50.6877 / 60) = 52.84479 deg E
```

These coordinates (36.44382, 52.84479) were entered into Google Maps, which confirmed the location as **Sari, Mazandaran Province, Iran**, consistent with the expected physical location of the test hardware. The map pin fell on a street-level position matching the test site, validating that the GNSS fix is accurate and correctly formatted.

Location Summary	
Coordinates (NMEA)	3626.6295 N, 05250.6877 E
Coordinates (decimal)	36.44382 deg N, 52.84479 deg E
Google Maps result	Sari, Mazandaran Province, Iran
Fix quality	1 (valid GPS fix)
Satellites in use	5
Altitude	46.2 m (HDOP 1.85 reading)

Figures 3 and 4 show the Claude.ai conversation used to decode and visualise the GPS output. The NMEA sentence was pasted in, and Claude identified the fix quality, decoded the coordinates, and displayed the location on a Google Maps embed, providing an independent confirmation of the fix.

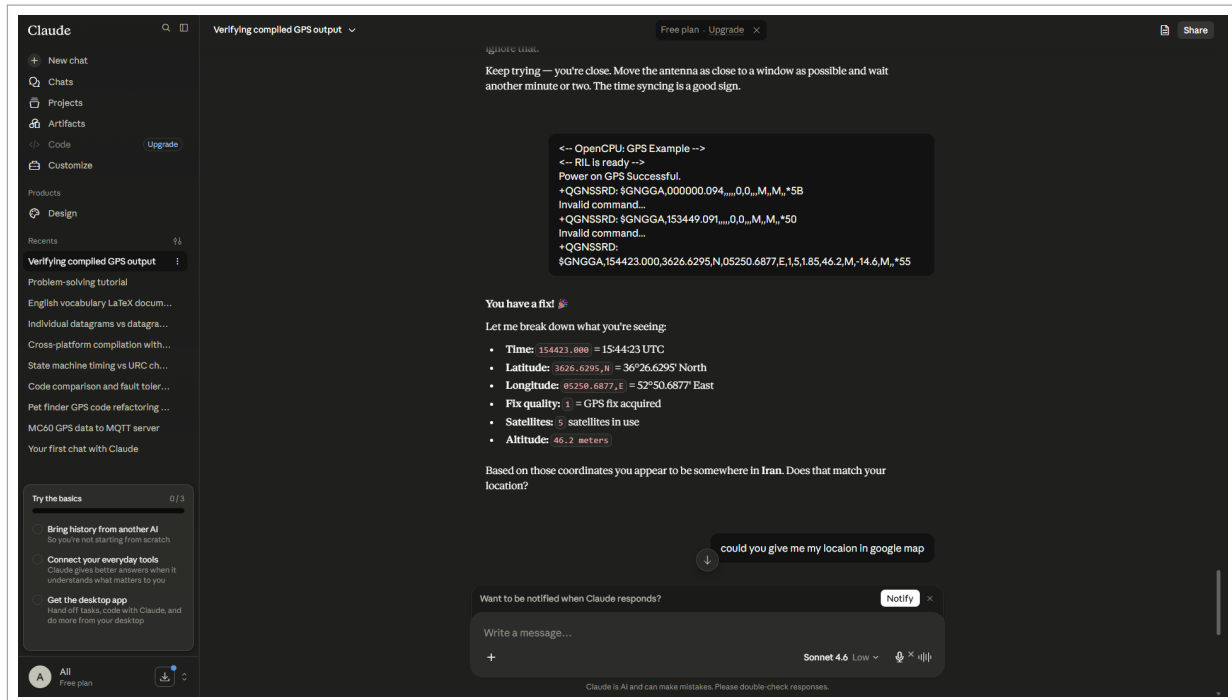


Figure 3: Claude.ai conversation decoding the \$GNGGA sentence — identifying the valid fix, satellite count, and coordinates from the raw NMEA output.

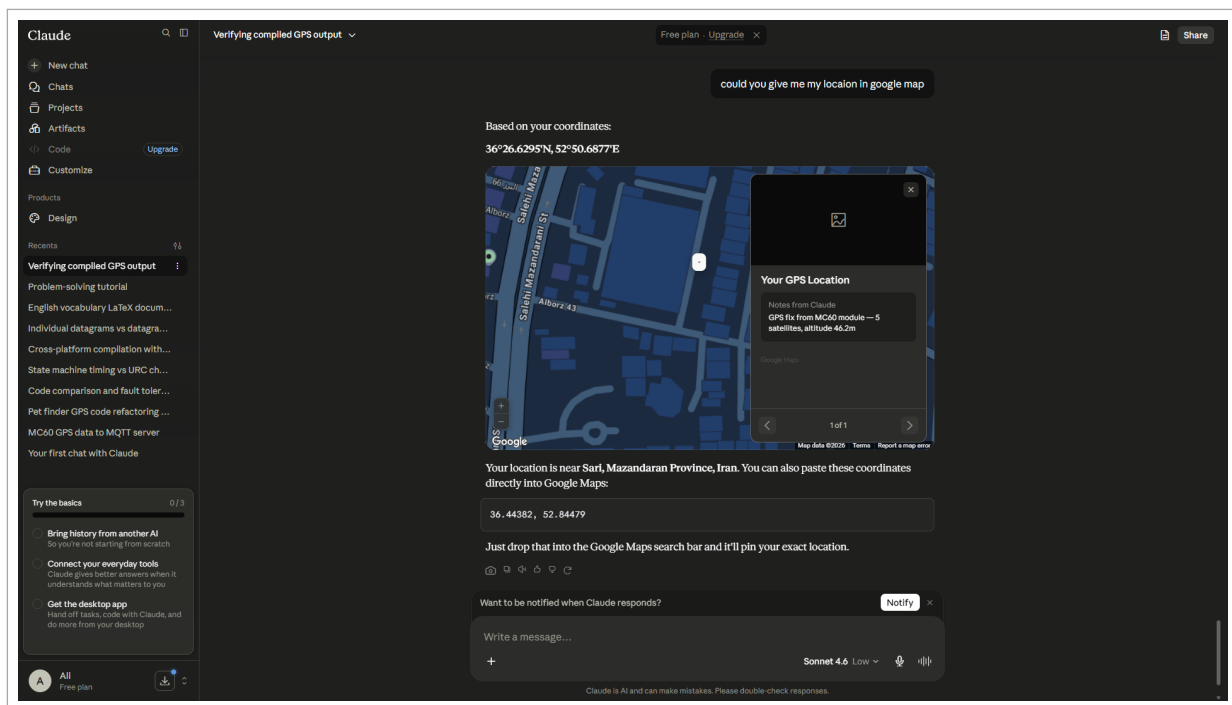


Figure 4: Google Maps embed in Claude.ai confirming the decoded coordinates (36.44382, 52.84479) correspond to Sari, Mazandaran Province, Iran.

6. Summary

Table 2 summarises the steps completed and the outcome of each.

Table 2: Test summary

Step	Notes	Status
Firmware (GPS)	Reused binary from first test — no recompile needed	Done
Flash	Not required — firmware already on module	Done
PuTTY monitor	Connected at 115200 baud on COM6	Done
Cold start wait	Pre-fix V-status sentences observed, then fix acquired	Done
Valid GNSS fix	2× \$GNGGA sentences, fix quality = 1, 4–5 satellites	Done
Coordinate decode	NMEA → 36.44382° N, 52.84479° E	Done
Location verify	Google Maps confirms Sari, Mazandaran Province, Iran	Done

Result

The third test is considered fully successful. The MC60's GNSS subsystem acquired a valid satellite fix after a short cold-start acquisition period. The decoded coordinates placed the device at the correct physical location in Sari, Iran, confirming that the GPS example firmware, the OpenCPU GNSS API, the external antenna, and the NMEA output pipeline are all working correctly end-to-end.

References

- [1] Quectel, *MC60 OpenCPU GS3 SDK V1.8*, SDK release package.
- [2] Arm Ltd., *Arm GNU Toolchain 15.2.1 (mingw-w64-i686-arm-none-eabi)*, <https://developer.arm.com/downloads/-/arm-gnu-toolchain-downloads>.
- [3] Silicon Labs, *CP210x USB to UART Bridge VCP Drivers*, <https://www.silabs.com/developers/usb-to-uart-bridge-vcp-drivers>.
- [4] First Test Report — MC60 OpenCPU GS3 SDK build and GPS example test (no fix), June 2026.
- [5] Second Test Report — MC60 OpenCPU GS3 SDK Bluetooth example test, June 2026.
- [6] NMEA 0183 Standard, National Marine Electronics Association, 2023.
- [7] *MC60_Build_Changes.pdf* — companion makefile fix documentation, June 2026.